FinSage

AI-Powered Financial Research Report Generator

You want to research **AAPL**. You have **two hours**, not two weeks. Analyst reports cost thousands. Free summaries are too shallow to trust. So **what do you actually do**?

WHY THIS, WHY NOW

30M+

New retail brokerage accounts opened since 2020 — a 3× surge in DIY investors

Source: SIFMA, FINRA 2024

<10%

Of US public companies get institutional research coverage — 90% of tickers are in the dark

Source: Bloomberg Research

100×

LLM inference cost drop in 24 months made agentic pipelines like ours economically viable

Source: a16z, 2025

THE OLD WAY VS THE FINSAGE WAY

THE MANUAL WAY

Research a ticker yourself

- ✕ Open 10+ tabs: Yahoo, SEC EDGAR, news sites
- ✕ Copy numbers into a spreadsheet by hand
- ✕ Skim a 200-page 10-K for red flags
- ✕ Build your own charts in Excel
- ✕ Rewrite notes when data updates tomorrow

Time: 4–8 hours · Cost: your evening

THE FINSAGE WAY

Click Generate, get a report

- ✓ 5 data sources pulled & cleaned automatically
- ✓ 4 AI agents collaborate on charts & analysis
- ✓ SEC filings parsed and summarized
- ✓ 8 refined charts with written interpretation
- ✓ Follow-up questions answered by built-in chatbot

Time: < 7 minutes · Cost: ~$2 in compute

WHAT HAPPENS WHEN YOU CLICK GENERATE

Live 5-stage multi-agent pipeline

~7 MIN END-TO-END

1 — 0:00 – 1:30
Fetch & Stage
Pulls prices, news, 10-K. dbt cleans into analytics.

2 — 1:30 – 3:30
Chart Agent
Draws 8 charts. AI vision critiques & regenerates.

3 — 3:30 – 4:30
Validation
Second AI verifies charts & numbers independently.

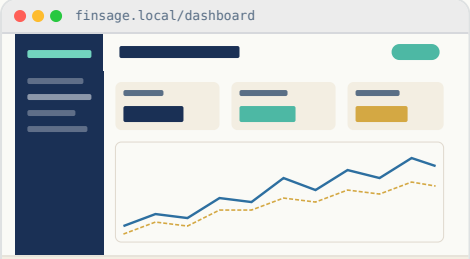
4 — 4:30 – 6:00
Analysis
Writes per-chart analysis & SEC summaries.

5 — 6:00 – 6:45
PDF Lands
Assembles 15–20 page branded PDF + chatbot.

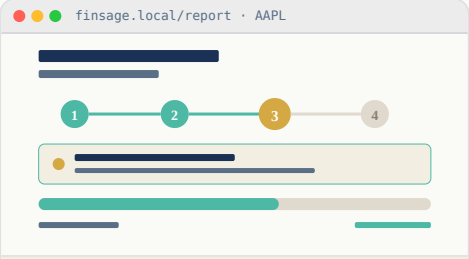
FinSage · Technical Deep Dive

Product · Architecture · Agents · Team

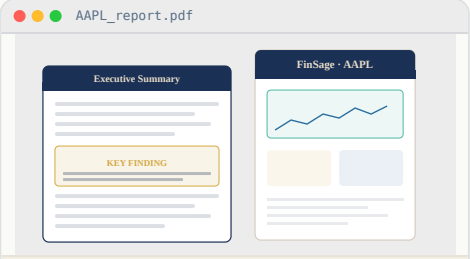
THE PRODUCT — WHAT USERS SEE



SCREEN 01
Dashboard — Portfolio View

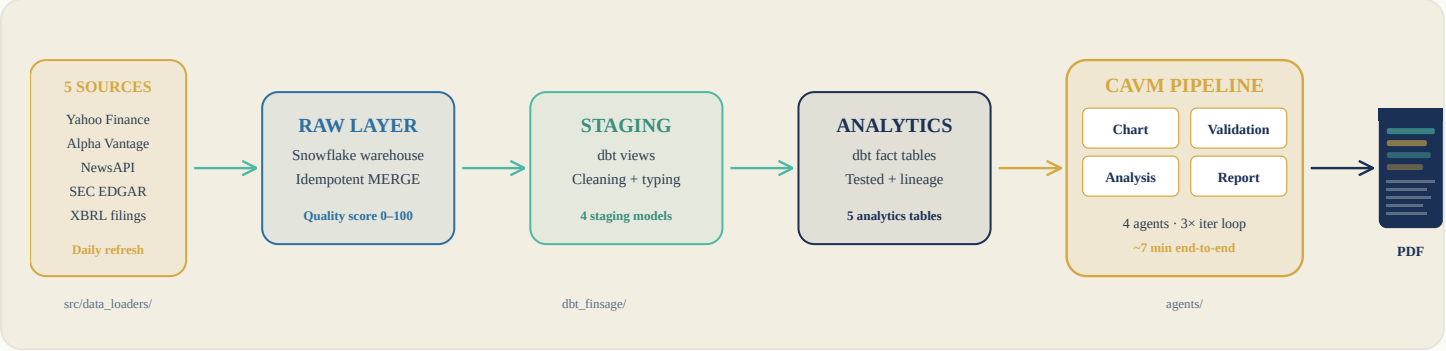


SCREEN 02
Pipeline — Live Agent Status



OUTPUT
Generated PDF — 15-20 pages

SYSTEM ARCHITECTURE — THREE-LAYER DATA FLOW



THE 4 CAVM AGENTS

AGENT 01 · CHART
Generate & self-refine
8 matplotlib charts from analytics tables. 3-iteration VLM refinement loop with Claude critiquing each chart until quality passes.
`claude-sonnet-4-6 · matplotlib · Cortex VLM`

AGENT 02 · VALIDATION
Independent quality check
Validates each chart for visual quality and underlying data integrity. Aborts pipeline if more than 2 charts fail validation.
`Cortex VLM · auto-retry × 3`

AGENT 03 · ANALYSIS
Per-chart written insight
LLM analysis tied to underlying numbers. Summarizes SEC MD&A and Risk Factors via Cortex SUMMARIZE. Structured JSON output.
`mistral-large · Cortex SUMMARIZE`

AGENT 04 · REPORT
Branded PDF assembly
15–20 page PDF with reportlab. Custom Midnight Teal theme: navy header, teal accents, gold highlights, color-coded signals.
`reportlab · Midnight Teal theme`

TECHNOLOGY STACK

Data Engineering
Snowflake · dbt 1.7 · Airflow 2.8 · Snowpark · Terraform · Docker

AI / LLMs
AWS Bedrock · Cortex LLM · Cortex VLM · Claude · Llama 3 · Mistral · Titan

Frontend & API
React 19 · Next.js 15 · FastAPI · Material UI · LangChain · reportlab

TEAM & HOW TO REACH US

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LIVE DEMO & SOURCE CODE

Try it yourself — or dive into the code

GITHUB `github.com/raghuneu/finsage`
DEMO `Pick a ticker → Generate → ask the chatbot`

SCAN TO VISIT